

chap6 数理统计的基本概念

一. 基本概念

数理统计: 样本信息 $\rightarrow$ 总体信息

1. 随机抽样、样本、样本空间、简单随机样本 (i.i.d.)

2. 经验分布函数

$$X_1, X_2, \dots, X_n \xrightarrow{\text{sort}} X_{(1)}, X_{(2)}, \dots, X_{(n)}$$

$$\Rightarrow F_n(x) = \begin{cases} 0, & x < X_{(1)} \\ \frac{k}{n}, & X_{(k)} \leq x < X_{(k+1)}, k=1, 2, \dots, n-1 \\ 1, & x \geq X_{(n)} \end{cases}$$

① 单调不减, 右连续

② $F_n(x) \xrightarrow{P} F(x) \quad (n \rightarrow +\infty)$ 伯努利定理

$$\text{实际上: } P\left(\sup_{x \in (-\infty, +\infty)} |F_n(x) - F(x)| \xrightarrow{n \rightarrow +\infty} 0\right) = 1$$

二. 统计量

1. 常见统计量 $\rightarrow$ 不依赖任何未知参数

$$\text{样本均值: } \bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$\text{样本方差: } S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

$$k\text{阶矩: } A_k = \frac{\sum_{i=1}^n X_i^k}{n}$$

$$k\text{阶中心矩: } B_k = \frac{\sum_{i=1}^n (X_i - \bar{X})^k}{n}$$

$$\max\{X_i\}, \frac{\bar{X}}{\sqrt{S^2}}, \dots$$

2. 统计量的无偏性

$$\textcircled{1} E(\bar{X}) = E(X)$$

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = E(X) \quad (X_i \text{ i.i.d. } X)$$

$$\textcircled{2} E(S^2) = D(X)$$

$$E(S^2) = \frac{1}{n-1} E\left(\sum_{i=1}^n (X_i - \bar{X})^2\right)$$

$$= \frac{1}{n-1} E\left[\sum_{i=1}^n [X_i - \mu - (\bar{X} - \mu)]^2\right]$$

$$= \frac{1}{n-1} E\left[\sum_{i=1}^n [(X_i - \mu)^2 - 2(X_i - \mu)(\bar{X} - \mu) + (\bar{X} - \mu)^2]\right]$$

$$= \frac{1}{n-1} E\left[\sum_{i=1}^n (X_i - \mu)^2 - 2n(\bar{X} - \mu)(\bar{X} - \mu) + n(\bar{X} - \mu)^2\right]$$

$$= \frac{1}{n-1} E\left[\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2\right]$$

$$= \frac{1}{n-1} (nD(X) - nD(\bar{X}))$$

$$= \frac{1}{n-1} (nD(X) - n \cdot \frac{1}{n} D(X))$$

$$= D(X)$$

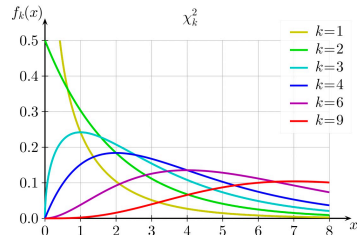
三. χ^2 分布

1. 定义

$$X_1, X_2, \dots, X_n \text{ i.i.d. } N(0, 1) \Rightarrow \chi_n^2 = \sum_{i=1}^n X_i^2 \sim \chi^2(n)$$

$$f_n(x) = \begin{cases} \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx$$



2. 性质

① χ^2 分布与其他分布的关系

$$\chi_1^2 = \Gamma\left(\frac{1}{2}, \frac{1}{2}\right) \quad \Gamma(\alpha, \lambda): f_X(x) = \begin{cases} \frac{x^{\alpha-1} \lambda^\alpha e^{-\lambda x}}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\chi_n^2 = \Gamma\left(\frac{n}{2}, \frac{1}{2}\right)$$

$$\chi_1^2 = \text{Exp}\left(\frac{1}{2}\right) \quad \text{Exp}(\lambda): f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\textcircled{2} E(\chi_n^2) = n, D(\chi_n^2) = 2n$$

$$\text{证: } E(\chi_n^2) = n E(X_j^2) = n \int_0^{+\infty} (t-i)^2 (e^{-\frac{t}{2}})^n \Big|_{t=0} = n$$

$$\begin{aligned} D(\chi_n^2) &= D\left(\sum_{i=1}^n X_i^2\right) = n \left(\overbrace{E(X_i^4)}^{\frac{3}{2}} - E^2(X_i^2) \right) \\ &= n \left((t-i)^4 (e^{-\frac{t}{2}})^n \Big|_{t=0} - 1 \right) \\ &= 2n \end{aligned}$$

$$\textcircled{3} \chi_n^2 + \chi_m^2 \sim \chi_{n+m}^2$$

$$\textcircled{4} n \text{ 很大 } (>40), \chi_{\alpha}^2(n) \approx \frac{1}{2} (2n + \sqrt{2n-1})^2$$

$\chi_{\alpha}^2(n)$ 的 α 上分位数

■ 例: 设总体 $X \sim N(\mu, \sigma^2)$, μ, σ^2 已知. $(X_1, X_2, \dots, X_n)$ 是取自总体 X 的样本, 求

(1) 统计量 $\chi^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$ 的分布;

△ (2) 设 $(n=5)$, 若 $a(X_1 - X_2)^2 + b(2X_3 - X_4 - X_5)^2 \sim \chi^2(k)$, 则 a, b, k 各为多少?

$$(1) \chi^2 \sim \chi^2(n)$$

$$(2) E(X_1 - X_2) = 0 \quad \text{由正态分布的可加性} \Rightarrow X_1 - X_2 \sim N(0, 2)$$

$$D(X_1 - X_2) = 2$$

$$\text{同理: } 2X_3 - X_4 - X_5 \sim N(0, 6)$$

$$\Rightarrow \begin{cases} \sqrt{a} = \sqrt{2} \\ \sqrt{b} = \sqrt{6} \end{cases} \Rightarrow \begin{cases} a = 2 \\ b = 6 \end{cases}$$

$$k = 2$$

■ 例: 已知 $X_i \sim \text{Exp}(\lambda)$ 且相互独立, 证明 $2\lambda \sum_{i=1}^n X_i \sim \chi^2(2n)$

$$\text{令 } Y_i = 2\lambda X_i, \text{ 则 } Y_i \sim \text{Exp}\left(\frac{1}{2}\right) = \chi^2(2)$$

$$\Rightarrow 2\lambda \sum_{i=1}^n X_i = \sum_{i=1}^n Y_i \sim \chi^2(2n) \quad \square$$

■ 例: 已知 $X \sim U(0, 1)$, 则有 $-2 \ln X \sim \chi^2(2)$

$$\text{令 } Y = -2 \ln X$$

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| = \begin{cases} \frac{1}{2} e^{-\frac{y}{2}}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

$$\Rightarrow Y \sim \text{Exp}\left(\frac{1}{2}\right) = \chi^2(2)$$

■ 例: 设总体 $X \sim N(\mu, \sigma^2)$, μ, σ^2 已知。 $(X_1, X_2, \dots, X_n)$ 是取自总体 X 的样本, 记 $\bar{X} = \sum_i X_i/n$, $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ 求

△(1) $\bar{X}$ 与 S^2 是否独立

(2) 统计量 $(n-1)S^2/\sigma^2$ 的分布;

证: (1) 令 $\vec{X} = (X_1, X_2, \dots, X_n)^T \sim N_n(\mu \vec{1}_n, \sigma^2 I_n)$

定义正交变换 $P: \vec{X} - \mu \vec{1}_n \rightarrow \vec{Y}$

其中

$$P_{n \times n} = \begin{bmatrix} \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \dots & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 & \dots & 0 & 0 \\ \frac{1}{\sqrt{2 \times 3}} & \frac{1}{\sqrt{2 \times 3}} & \frac{-2}{\sqrt{2 \times 3}} & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \dots & \frac{1}{\sqrt{(n-1)n}} & \frac{-(n-1)}{\sqrt{(n-1)n}} \end{bmatrix} \quad (PP^T = I_n)$$

P 是正交变换 $\Rightarrow \vec{Y} \sim N(0, \sigma^2 PP^T) = N(0, \sigma^2 I_n)$

$\Rightarrow Y_1, Y_2, \dots, Y_n$ 两两不相关

$\Leftrightarrow Y_1, Y_2, \dots, Y_n$ 两两独立 (正态分布)

$$Y_1 = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) = \sqrt{n}(\bar{X} - \mu) \Rightarrow \bar{X} = \frac{Y_1}{\sqrt{n}} + \mu$$

$$\begin{aligned} \sum_{i=2}^n Y_i^2 &= \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X} + \bar{X} - \mu)^2 - n(\bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2 \sum_{i=1}^n (X_i - \bar{X})(\bar{X} - \mu) + n(\bar{X} - \mu)^2 - n(\bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2 \left(\sum_{i=1}^n X_i - n\bar{X} \right) (\bar{X} - \mu) \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 \end{aligned}$$

$$\Rightarrow S^2 = \frac{1}{n-1} \sum_{i=2}^n Y_i^2$$

故 $\bar{X}$ 与 S^2 独立. □

(2) $\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n \tilde{X}_i^2 = \chi^2(n)$ ✗

不是 μ !

$= \frac{1}{\sigma^2} \sum_{i=2}^n Y_i^2 = \chi^2(n-1)$

($Y_i \sim N(0, \sigma^2)$)

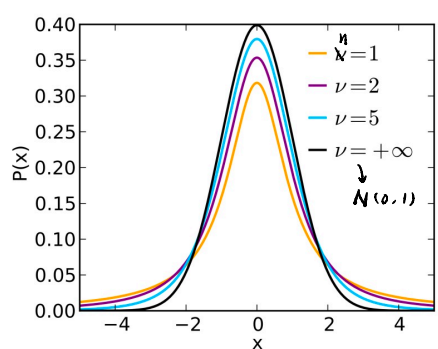
四. t 分布

1. 定义

X, Y 相互独立, $X \sim N(0, 1), Y \sim \chi^2(n)$

$$\Rightarrow T = \frac{X}{\sqrt{\frac{Y}{n}}} \sim t(n)$$

$$f_T(t, n) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi} \Gamma(\frac{n}{2})} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}, \quad t \in (-\infty, +\infty)$$



2. 性质

① n 很大时 (>45), $t_\alpha(n) \approx Z_\alpha$

■ 例: 设总体 $X \sim N(\mu, \sigma^2)$, μ, σ^2 已知。 $(X_1, X_2, \dots, X_n)$ 是取自总体 X 的样本, 记 $\bar{X} = \sum_i X_i/n$, $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ 求 $\frac{\bar{X} - \mu}{\sqrt{S^2/n}}$ 的分布

解: $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$(n-1)S^2 \sim \chi^2(n-1)$$

$$\Rightarrow \frac{\bar{X} - \mu}{\sqrt{\frac{S^2}{n}}} = \frac{\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}}{\sqrt{\frac{(n-1)S^2/\sigma^2}{(n-1)}}} \sim \frac{N(0, 1)}{\sqrt{\chi^2(n-1)/n}} \sim t(n-1)$$

■ 设 $X_1, X_2, \dots, X_{n_1}$ 为取自正态总体 $N(\mu_1, \sigma^2)$ 的样本, $Y_1, Y_2, \dots, Y_{n_2}$ 为取自正态总体 $N(\mu_2, \sigma^2)$ 的样本, 并且这两个样本相互独立, 则随机变量 $t = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ 服从自由度为 $n_1 + n_2 - 2$

的 t 分布, 其中 $\bar{X}$ 与 $\bar{Y}$ 分别为这两个样本的样本均值,

$$S_w^2 = \frac{\sum_i (X_i - \bar{X})^2 + \sum_j (Y_j - \bar{Y})^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

证: $\bar{X} - \bar{Y} - \mu_1 + \mu_2 \sim N(0, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2})$

$$S_w^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2 \sim \chi^2(n_1 + n_2 - 2)$$

$$\Rightarrow t = \left(\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \right) \bigg/ \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{\sigma^2}} \bigg/ \sqrt{\frac{1}{n_1 + n_2 - 2}}$$

$$\sim t(n_1 + n_2 - 2).$$

□

五. F分布. ■ 设总体 $X \sim N(\mu, \sigma^2)$, $(X_1, \dots, X_4)$ 与 $(Y_1, \dots, Y_9)$ 是取自总体 X 的两个独立样本, $\bar{X}, S_1^2$ 和 $\bar{Y}, S_2^2$ 分别为样本均值和样本方差; 求 (1) $D(S_1^2 - S_2^2)$; (2) $a \frac{\bar{X} - \bar{Y}}{S_1^2} \sim t(k)$, 则 a, k 各为多少? (3) $b \sum_{i=1}^4 (X_i - \mu)^2 / S_2^2 \sim F(n_1, n_2)$, 则 b, n_1, n_2 各为多少?

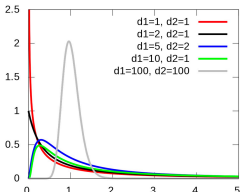
1. 定义

$X \sim \chi^2(n_1), Y \sim \chi^2(n_2), X, Y$ 相互独立

$$F = \frac{\frac{X}{n_1}}{\frac{Y}{n_2}} \sim F(n_1, n_2)$$

$$f(x; n_1, n_2) = \begin{cases} \frac{1}{B(\frac{n_1}{2}, \frac{n_2}{2})} n_1^{\frac{n_1}{2}} n_2^{\frac{n_2}{2}} x^{\frac{n_1}{2}-1} (n_2 + n_1 x)^{-\frac{n_1+n_2}{2}} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = \frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}$$



2. 性质

① $F \sim F(n_1, n_2) \Rightarrow F^{-1} \sim F(n_2, n_1)$

$$F = \frac{\frac{X}{n_1}}{\frac{Y}{n_2}}, F^{-1} = \frac{Y/n_2}{X/n_1} \sim F(n_2, n_1)$$

② $X \sim t(n), X^2 \sim F(1, n)$

$$X = \frac{Y}{\sqrt{\frac{Z}{n}}}, Y \sim N(0, 1), Z \sim \chi^2(n)$$

$$X^2 = \frac{Y^2}{\frac{Z}{n}} \sim F(1, n)$$

③ $F_{1-\alpha}(n_1, n_2) = \frac{1}{F_{\alpha}(n_2, n_1)}$

$$P(F > F_{1-\alpha}(n_1, n_2)) = 1 - \alpha$$

$$P(F > \frac{1}{F_{\alpha}(n_2, n_1)}) = P(F^{-1} < F_{\alpha}(n_2, n_1))$$

$$= P(F^{-1} < F_{\alpha}(n_2, n_1)) \sim F(n_2, n_1)$$

$$= 1 - \alpha$$

□

■ 例: 设总体 $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2), \mu_1, \sigma_1^2, \mu_2, \sigma_2^2$ 已知. $(X_1, X_2, \dots, X_{n_1})$ 是取自总体 X 的样本, $(Y_1, Y_2, \dots, Y_{n_2})$ 是取自总体 Y 的样本, 记 $\bar{X} = \sum_{i=1}^{n_1} X_i / n_1, S_X^2 = \frac{1}{n_1-1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2, \bar{Y} = \sum_{i=1}^{n_2} Y_i / n_2, S_Y^2 = \frac{1}{n_2-1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$ 求 $F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$ 的分布

$$\frac{(n_1-1) S_X^2}{\sigma_1^2} \sim \chi^2(n_1-1)$$

$$\frac{(n_2-1) S_Y^2}{\sigma_2^2} \sim \chi^2(n_2-1)$$

$$\Rightarrow F = \frac{\frac{\chi_{n_1-1}^2}{n_1-1}}{\frac{\chi_{n_2-1}^2}{n_2-1}} \sim F(n_1-1, n_2-1)$$

$$\bar{X} = \frac{1}{4} \sum_{i=1}^4 X_i, \bar{Y} = \frac{1}{9} \sum_{i=1}^9 Y_i$$

$$S_1^2 = \frac{1}{3} \sum_{i=1}^4 (X_i - \bar{X})^2, S_2^2 = \frac{1}{8} \sum_{i=1}^9 (Y_i - \bar{Y})^2$$

(1) $\frac{3 S_1^2}{\sigma^2} \sim \chi^2(3), \frac{8 S_2^2}{\sigma^2} \sim \chi^2(8)$

$$D(S_1^2 - S_2^2) = D(S_1^2) + D(S_2^2)$$

$$= D(\frac{\sigma^2}{3} \chi_3^2) + D(\frac{\sigma^2}{8} \chi_8^2)$$

$$= \frac{\sigma^4}{9} \times 3 + \frac{\sigma^4}{64} \times 16$$

$$= \frac{2\sigma^4}{3} + \frac{\sigma^4}{4}$$

$$= \frac{11}{12} \sigma^4$$

(2) $S_1 = \sqrt{\frac{\sigma^2 \chi_3^2}{3}}$

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{4}), \bar{Y} \sim N(\mu, \frac{\sigma^2}{9})$$

$$\Rightarrow \bar{X} - \bar{Y} \sim N(0, \frac{13}{36} \sigma^2)$$

$$\Rightarrow \frac{\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{13}{36} \sigma^2}}}{\frac{S_1}{\sigma}} \sim t(3)$$

$$\Rightarrow \alpha = \frac{b}{\sqrt{13}}, k = 3$$

(3) $\frac{\sum_{i=1}^4 (X_i - \mu)^2}{\sigma^2} \sim \chi^2(4)$

$$\Rightarrow \frac{\frac{\sum_{i=1}^4 (X_i - \mu)^2}{4\sigma^2}}{\frac{8 S_2^2}{8\sigma^2}} \sim F(4, 8)$$

$$\Rightarrow b = \frac{1}{4}, n_1 = 4, n_2 = 8$$